



Aerodynamics Numerical Project

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Introduction

In this project, we will simulate the flow around a NACA 4-digit airfoil. The first digit represents the percentage of camber in relation to the chord, the third digit represents the position of the maximum camber in 1/10th of the chord, and the last two digits represent the thickness in percentage of the chord.

In the first part, we will use the FreeFem solver to simulate the flow past the NACA airfoil. We will calculate both the inviscid and viscous flow using a numerical solution of Laplace's equation for the stream function.

$$\Delta(\psi) = 0 \quad (1)$$

Furthermore, we will investigate why the Kutta condition at the rear is crucial in determining the flow circulation around the airfoil.

In the second part, we will implement the theory of thin profiles in a Python code to assess the validity of the FreeFem simulation and compare its results to the theoretical results.

In the final part, we will examine the impact of the tandem effect on the lift and circulation of the wing.

1 Single airfoil simulation

In this part, we used the code `naca.edp` that calculates the inviscid flow past a NACA airfoil. The goal was to assess the validity of the potential flow theory on incompressible flows and determine the angle of zero lift for different airfoil designs (cf Annex: Single Airfoil).

1.1 The Kutta condition

The Kutta condition imply that the flow from the topside and underside of a surface join up at the sharp trailing edge and leave it tangentially.

For an airfoil, the Kutta condition is met when the rear stagnation point is located at the trailing edge TE.

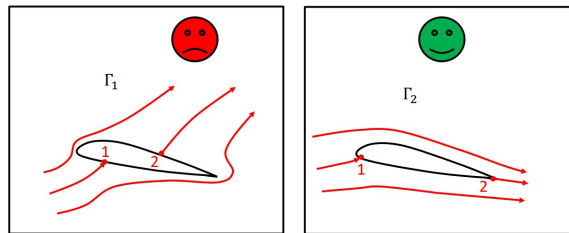


Figure 1: Sketch of circulation such that Kutta condition is satisfied

To impose this condition, a large-scale circulation is combined with the potential flow. Which can be expressed by means of stream function as :

$$\psi = \psi_0 + \beta \cdot \psi_1 \quad (2)$$

β is adjusted to comply with the above condition

In the case of small camber, it can be given by:

$$\beta = -\frac{\psi_0(x^+, y^+ + \epsilon_y) + \psi_0(x^-, y^- - \epsilon_y)}{\psi_1(x^+, y^+ + \epsilon_y) + \psi_1(x^-, y^- - \epsilon_y) - 2} \quad (3)$$

Where : $+$, $-$ denotes respectively the upper and lower surfaces at the vicinity of TE.

Nonetheless, Kutta condition is a consequence of the numerical constraint of potential theory. In reality at the trailing edge is never perfectly sharp and viscosity cancels the flow velocity at the surface. This removes the constraint of tangential flow continuity at the trailing edge and allows flow around the TE for short transient.

Simulation: We ran simulations on a symmetric NACA airfoil 0310 at an angle of attack 0 and 20° respectively, using the script "naca.edp".

By setting the adjustment coefficient β , to zero. We suppress the kutta condition. So that both cases with & without Kutta condition implementation would be examined.

We retrieve from the code generated output the values of the coefficients: β , c_d and c_l , the latter will be compared to theory. Using the formula from potential theory, we get:

$$c_l = 2\pi \sin(\alpha) \quad (4)$$

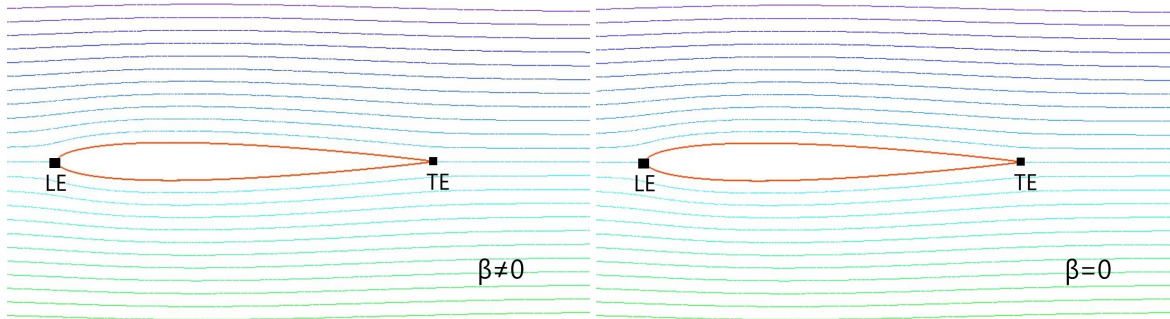


Figure 2: Streamlines around the NACA 0310 Airfoil at $\alpha = 0$.

From Figure 2 :

- The streamlines are parallel downstream and leaves the stagnation point that is located at the TE smoothly;
- The Kutta condition is than satisfied in both cases for an angle of attack $\alpha = 0$;
- The outputted values of the lift coefficient are: $c_{l,num} = -8.1 \times 10^{-5}$ (with K.C), $c_{l,num} = -1.3 \times 10^{-4}$ (without K.C), whereas the theoritical value $c_{l,th} = 0$

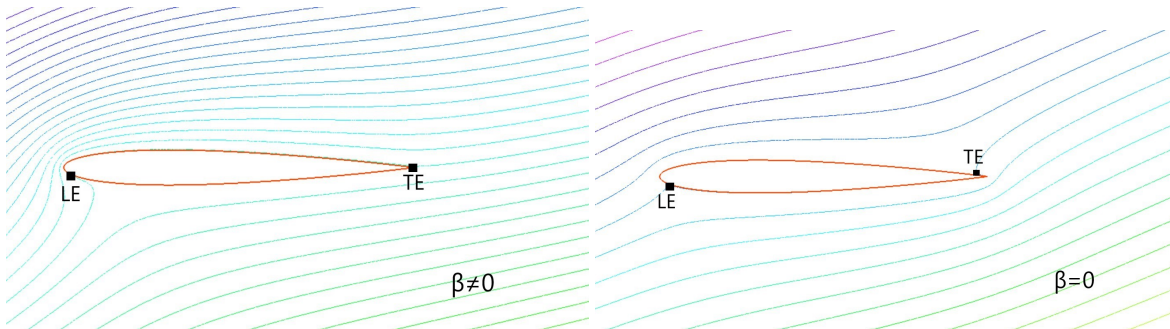


Figure 3: Streamlines around the NACA 0310 Airfoil at $\alpha = 20^\circ$.

From Figure 3 :

- when $\beta = 0$, Flow detaches from the upper surface at the topside stagnation point. Underside flow goes around TE towards the topside stagnation point. With K.C , the topside stagnation point is moved to the TE, there is no flow around the latter.
- The Kutta condition is satisfied only for $\beta \neq 0$ for an angle of attack $\alpha = 20^\circ$;
- The outputted values of the lift coefficient are: $c_{l,num} = 2.31$ (with K.C), $c_{l,num} = -0.11$ (without K.C) whereas the theoritical value $c_{l,th} = 2.15$

Using PARAVIEW (figure 4) we observed: for (a) the same pressure on the upper and the lower surfaces induces no lift however, an incidence produces lift by creating a pressure gradient (b).

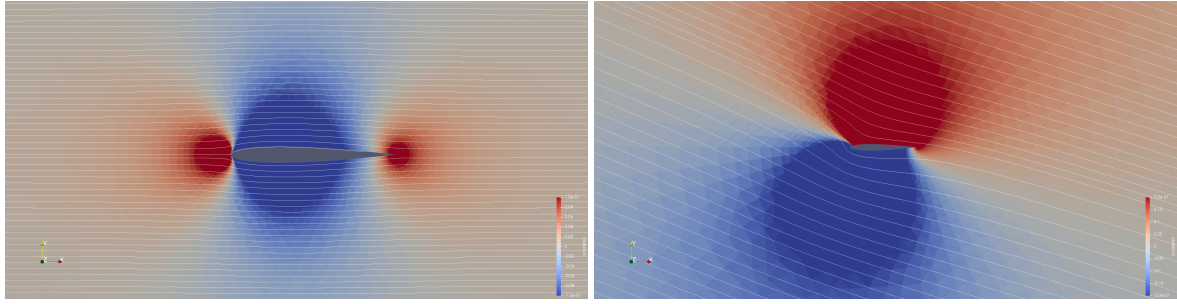


Figure 4: Kutta Condition, pressure distribution around NACA 0310 , (a) angle $\alpha = 0$ and (b) angle $\alpha = 20^\circ$

1.2 The angle of zero lift

We choose a symmetric airfoil NACA 0310 and a cambered one NACA 4310 to study the variation of the lift coefficient under several angles of attack, which can be done iteratively using the script "angle_of_attack_polar_potential.edp". In a further step, we carry a similar study in a viscous flow by assigning a finite and small value to the Reynolds number using "naca_viscous.edp".

1.2.1 Inviscid flow

The law of lift of the cambered and symmetric airfoils are plotted in the same graph (figure 5), together with the flat plate law $k_{flatplate} = \pi$. The symmetric airfoil law is piled up on the linear law for α in the range of $[-5, 5]$, then the approximation of small angles is not valid as the two laws diverge.

The lift gradient : is the slope of the lift coefficient ,it can be expressed as:

$$k = \frac{1}{2} \frac{dc_l}{d\alpha} \quad (5)$$

We find that lift slope is independent of the choice of camber such that $k_{symmetric} = k_{cambered} = 3.34$ which are close to the flat plate value $k_{flatplate} = 3.14$

A symmetric airfoil is shown to have an angle of zero lift equal to 0° as might be expected. The camber introduces a shift in the lift law that allows us to retrieve the angle of zero lift.

We noticed that the shift became larger as the camber is increased. A low camber airfoil (4 percent camber) undergoes zero lifts at $\alpha_0 = -3.8^\circ$, while a wing with 8 percent camber has an angle of zero lift equals to -7.9° .

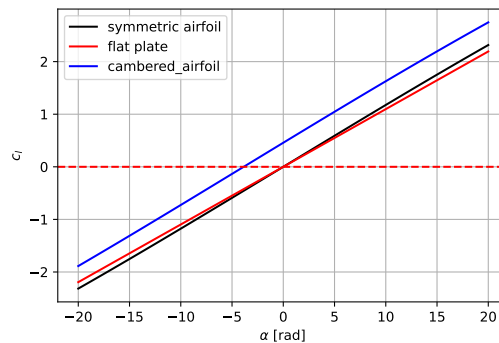


Figure 5: Coefficient of lift as a function of angle of attack, inviscid flow

1.2.2 Viscous flow

Unlike the inviscid assumption, the law of lift in viscous flow fits a non-linear curve. The behaviour of the angle of zero lifts is the counterpart of the previous case, however, the asymmetric (4 percent camber) airfoil's

angle of zero lift is significantly decreased to -1.3° .

We see that the lift coefficient evolves alike in symmetric and cambered airfoils approaching the typical lift curve with a stall delay. The stall occurs at critical values of the angle of attack about 30° .

Indeed, at too low Re , a wing may stall at a higher angle of attack and reaches lower lift coefficient.

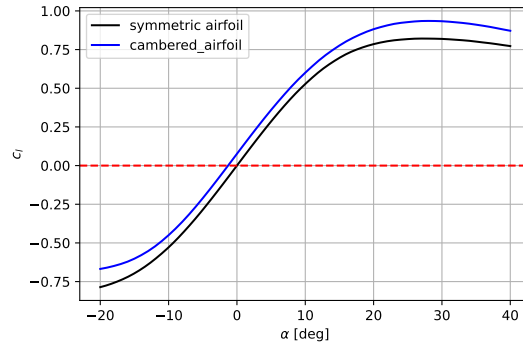


Figure 6: Coefficient of lift as a function of angle of attack for $Re = 100$

2 Thin airfoil theory

The Thin Airfoil Theory is a mathematical model used to predict the lift and drag characteristics of an airfoil, which is the shape of an aircraft wing. The theory is based on the assumption that the airfoil is thin, meaning the ratio of its thickness to its chord is small.

The theory uses linearized potential flow to calculate the lift and drag on the airfoil. The airfoil is considered to be a two-dimensional object, and the flow around it is assumed to be *inviscid* and *irrotational*. The flow is also assumed to be *incompressible*, and to have a constant speed far from the airfoil.

The lift force on the airfoil is calculated using the Kutta-Joukowski theorem, which states that the lift is equal to the circulation of the flow around the airfoil times the fluid density. The circulation is calculated using the velocity potential, which satisfies the Laplace equation and the Kutta condition at the trailing edge of the airfoil. The drag force is calculated using the pressure distribution on the airfoil, which is found by applying the Bernoulli equation to the flow.(cf: Annex)

FreeFem solve the Laplace problem directly, whereas python solve only the airfoil theory. We trace the results using the python code "thin_airfoil_naca.py"

2.1 Effect of incidence on the lift

We first carried out a study for an airfoil without camber and angle of attack (figure 7). We could observe that there was no lift around the wing profile. The results between the simulation and the theory are identical.

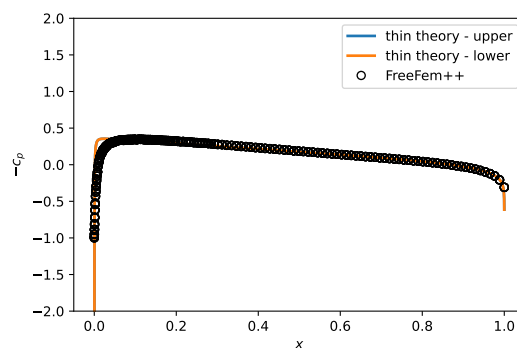


Figure 7: Comparison of FreeFem and python for NACA 0310 airfoil, without incidence

We then made a study for an aerodynamic profile with a slight camber and an angle of attack (figure 8). We could observe the lift distribution around the profile. The results between simulation and theory are almost identical. The theory of thin airfoils is not defined in 0. So the results for x are very close to 0 but not exact.

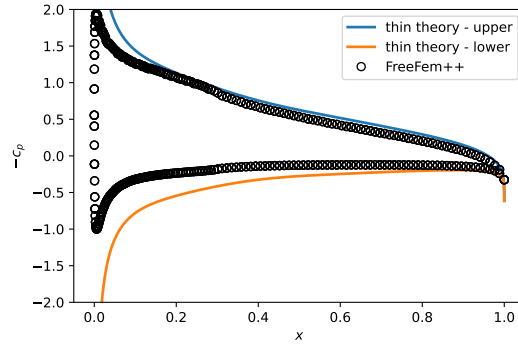


Figure 8: Comparison of FreeFem and python for NACA 2310 airfoil, with incidence

2.2 Validity of the thin airfoil theory

2.2.1 Different angle of attack

We tried to discuss the validity of the theory by varying the angle of attack of the wing.

In the figure ??, we can see that the simulation and the theory diverge when the α increases. Indeed, we expected this result. The critical angle is located between 5 and 10.

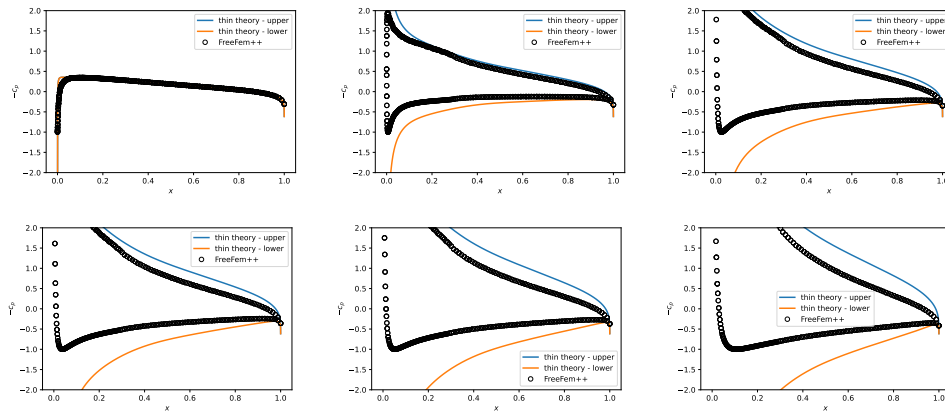


Figure 9: Comparison of the pressure distribution between the python code and the simulation when the α angle increases, $\alpha=[0,5,10,12,15,20]$

2.2.2 Different camber

We tried to discuss the validity of the theory by varying the angle of attack of the wing. In figure 10, we can see that the simulation and the theory diverge when the camber increases. Indeed, in the thin airfoil theory, the approximation is that the camber of the wing is negligible compared to the chord. When the camber becomes too high, this approximation is no longer valid.

Visually we have estimated that the critical camber is 4. We tried to quantify this error with a python code. We compared the maximum difference between the two curves. We found that the gap increases considerably from camber=4.

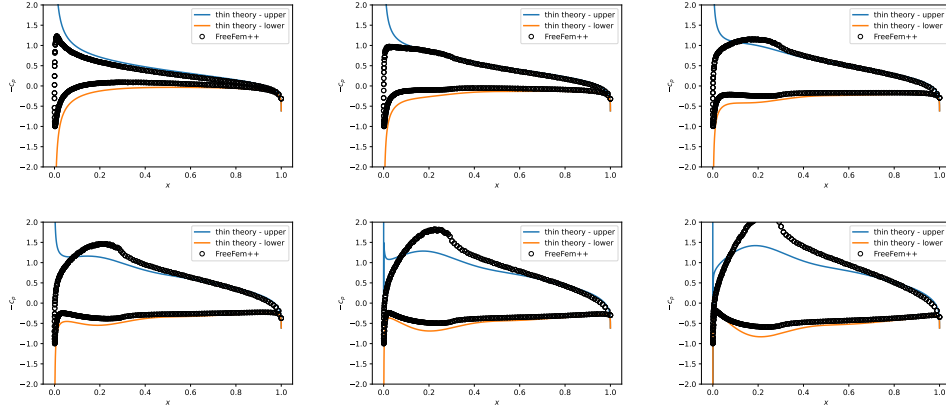


Figure 10: Comparison of the pressure distribution between the python code and the simulation when the camber increases, camber = [0,2,4,6,8,10]

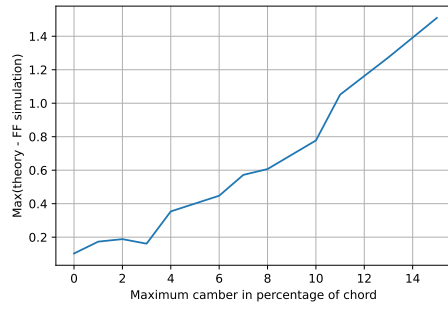


Figure 11: Graph of the maximum difference between theory and simulation as a function of camber

2.3 Different thickness

Finally in the same way we varied the thickness of the wing. In the figures below we have seen that the theory is no longer valid for starting from the thickness of a NACA2318 profile, which was estimated visually.

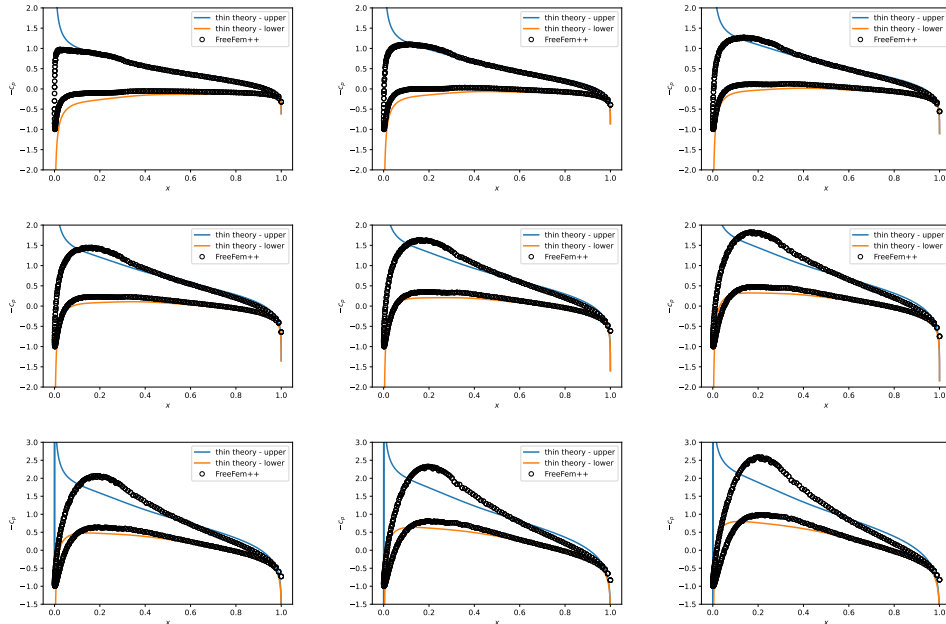


Figure 12: Comparison of the pressure distribution between the python code and the simulation when the thickness increases, NACA23[10,14,18,22,26,30,35,40,45]

3 Tandem Airfoils

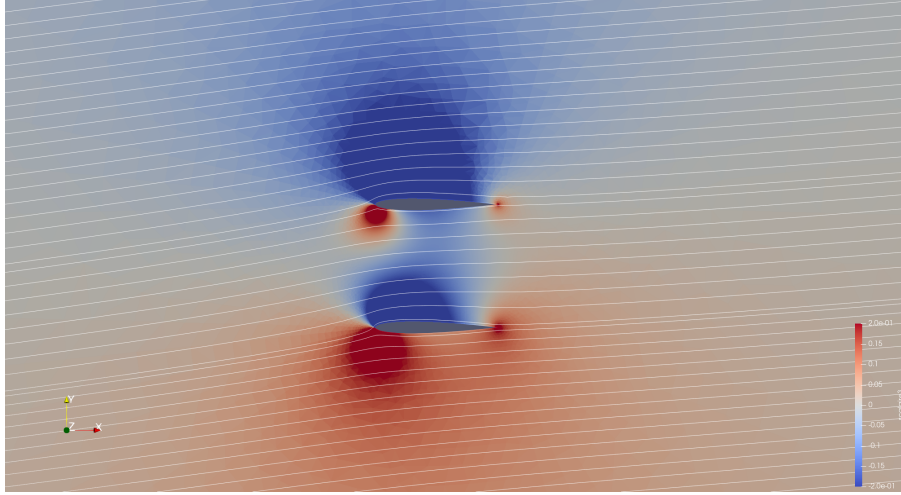


Figure 13: Pressure and streamline for the Tandem, $\alpha = 6$ [deg]

The 2D tandem airfoil theory in potential flow describes the flow around two airfoils placed closely together. The theory assumes that the flow is inviscid and irrotational, and that the airfoils are thin.

In the cases of the mono-plan, simple airfoil: The lift and drag coefficients of the tandem airfoils can be determined using the following equations:

$$C_l = \frac{2}{\rho V_\infty^2} \int_0^c \left(\frac{\partial \phi}{\partial y} \right)_{x=c} dx$$

$$C_d = \frac{2}{\rho V_\infty^2} \int_0^c \left(\frac{\partial \phi}{\partial x} \right)_{y=0} dx$$

Where ϕ is the velocity potential, ρ is the density of the fluid, V_∞ is the freestream velocity, c is the chord length of the airfoils, x and y are the coordinates in the plane of the airfoils, and the subscripts indicate that the derivatives are taken at the respective locations.

One key aspect of the 2D tandem airfoil theory is the use of the concept of circulation, which is defined as the line integral of the velocity potential along a closed contour. The circulation around the tandem airfoils is used to calculate the lift force acting on the airfoils. Additionally, the Kutta-Joukowski theorem is also applied. The theory has certain limitations. We assumed *inviscid* and *irrotational* flow, without turbulence...

We used a python code from a project that we did last years, to compare the solution with direct solving Laplace equation and the theory of the bi-plan that we will explain. We first consider a single element figure which consists of a segment $P_j Q_j$, on which are placed:

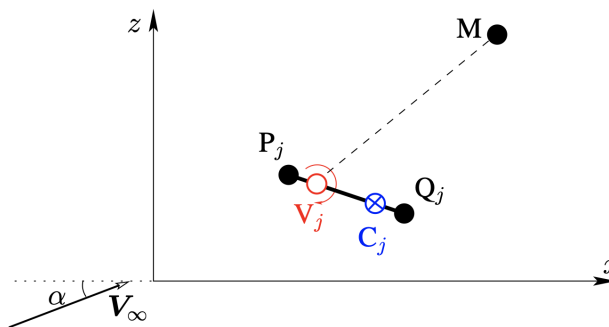


Figure 14: Illustration for a mono-plan

- a point circulation vortex Γ_j at point V_j located at a quarter of the segment,
- control point C_j located at three quarters of the segment.

The python code solves this equation 12 (Cf: Annex, Tandem airfoils theory).

3.1 Coefficient of lift as a function of α

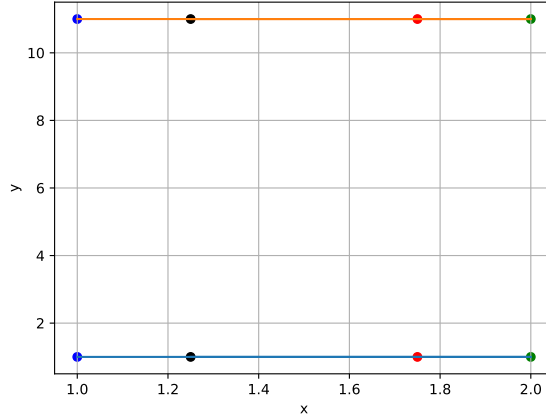


Figure 15: Tandem representation computed with python

In theory the lift should be identical on the profiles of the wings, this is what we see on the theoretical results of the python code. We then carried out simulations with symmetrical naca profiles 0310. Spaced by a distance $d=1$. The variation of the angle of attack of the two profiles showed that the lift increases linearly with the angle of attack, at least for low angles of attack. (figure 16) We found that the theories performed on python and the measurements performed on FreeFem follow the same behavior.

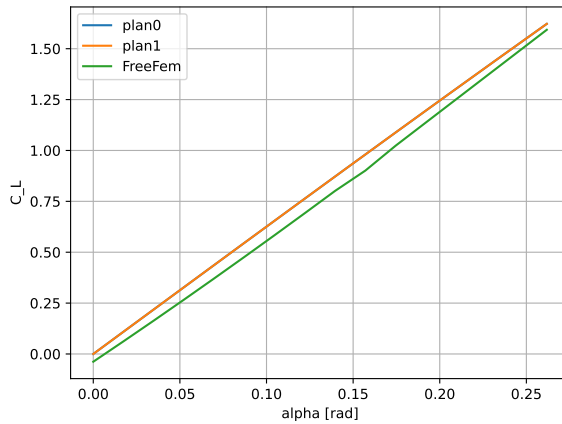


Figure 16: Comparison of lift coefficient as a function of α for FreeFem simulation and theory calculated with python

A slight difference was observed, this may be due to the difference between the FreeFem wing thickness. Indeed, the python code assumes that the thickness is negligible.

3.2 Coefficient of lift as a function of distance d

In theory when we vary the distance, the lift on a wing profile should tend towards a constant value. This is what we saw on the theoretical results of the python code.

- If $d \rightarrow 0$ (d, the distance between the two plates), : $\Gamma_1 = \Gamma_2 = \frac{\Gamma_0}{2}$
- If $d \rightarrow \infty$ (d, the distance between the two plates), : $\Gamma_1 = \Gamma_2 = \Gamma_0$
- When there are two plates, by symmetry, we have $\Gamma_1 = \Gamma_2$; and therefore by proportionality ($\propto 2\pi\alpha$) the lift coefficients are equal. In other words, when we plot lift as a function of α and d, the curves associated with each plate (and each Γ) overlap.

We then performed simulations with 0310 naca profiles symmetrical with a slight 3° angle. The variation of the distance between the two profiles (figure 18) showed that the lift increases then tends towards a constant value. We found that the theories performed on python and the measurements performed on FreeFem follow the same

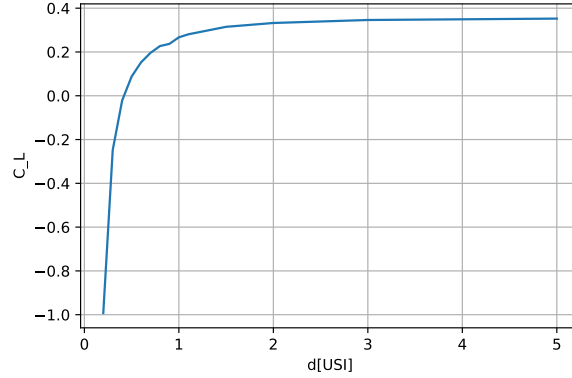


Figure 17: Lift coefficient as a function of distance d with FreeFem

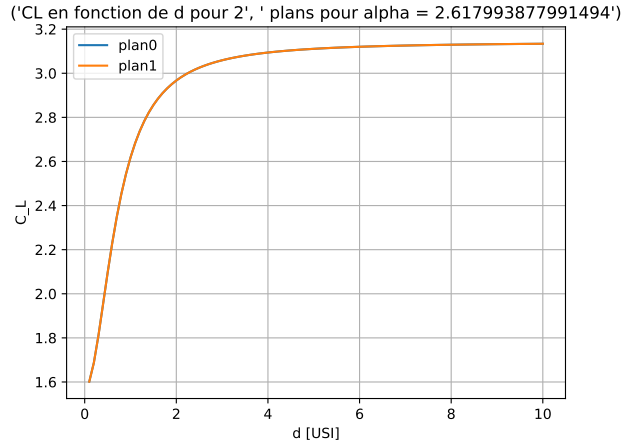


Figure 18: Lift coefficient as a function of distance d for $alpha = 15$ [deg] with python

behavior.(Figure 16 and Figure 18). However the values are not the same, this is due to the difference in the implementation of the data (the angle α) between the two. We have therefore carried out a study here on the lift as a function of distance and angle of inclination of the wings in the case of the Tandem. We can see that the addition of a second wing adds lift. However the implementation of such a system implies other constraints that we will not deal with here.

4 Conclusion

In this project, we studied the lift produced by an airfoil in different situations. Therefore, we used different airfoil designs, with different camber, thickness and also incidence.

The first step was to check the flow at the trailing edge by implementing the Kutta Condition given the sensitivity of the solution to the good application of this condition.

Using the potential flow theory, one of the first things noticed is the fact that in a cambered airfoil, one must move to a negative angle of attack to obtain zero lift coefficient (hence zero lift). This angle is called the angle of zero lift.

The second step was to validate the thin airfoil theory, and check that this theory is in good agreement with FreeFem Simulations only for a certain range of incidence, thickness and camber. The last investigation was the case of Tandem airfoils. We studied the effect of the incidence on the two airfoils and distance between them on the lift. Ultimately we compared this result with a python code of the circulation theory.

Note that, in this project we focused on the lift rather than the drag. For instance in the case of Tandem, it's true that the lift increase but at the cost of additional drag.

5 Annex

5.1 Single Aifoils, theory

The inciscid flow past NACA airfoil are simulate on the basis of the potential flow theory. The theory of potential flow around an airfoil is based on the assumption that the flow is *irrotational* and *incompressible*:

$$\nabla \cdot \vec{u} \text{ and } \vec{\nabla} \times \vec{u}$$

. Physically the fluid's movement is described by streamlines that do not cross and whose volumetric flow rate remains constant.

The flow is assumed here to be inviscid, the Navier-Stokes equation used to describe these flows can therefore be expressed mathematically as follows:

$$\frac{\partial}{\partial x}(u^2 + v^2) + \frac{\partial}{\partial y}(uv) = -\frac{1}{\rho} \frac{\partial p}{\partial x} \quad (6)$$

where u and v are the velocity components in the x and y directions, p is the pressure, and ρ is the fluid density. Using the irrotationality assumption, it can be deduced that the stream function, Φ , is a potential function, i.e.

$$u = \frac{\partial \Phi}{\partial y} \text{ and } v = -\frac{\partial \Phi}{\partial x}$$

Using these relations, we can then solve for Φ in terms of p . Using this solution for Φ , we can then calculate the velocity components in the x and y directions, and thus describe the flow around an airfoil.

From the conservation of mass and the Cauchy-Reiman condition, we can say that:

$$\begin{aligned} \frac{\partial}{\partial x} u &= \frac{\partial}{\partial y} v \\ \frac{\partial}{\partial y} u &= -\frac{\partial}{\partial x} v \end{aligned}$$

That is to say that we can define Φ and Ψ are harmonic function. The complex potential function is therefore defined as

$$f(x) = \Psi(x, y) + i\Phi(x, y) \quad (7)$$

The advantage of defining such a holomorphic function is the possibility of describing the current field as a sum of elementary solution. The viscosity of the fluid is neglected here, we study later the effect of viscosity and the bondary layer.

5.2 Thin airfoil theory

The equation for the velocity potential of the flow around a thin airfoil is given by

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0 \quad (8)$$

The circulation around the airfoil is given by

$$\Gamma = \int_C \mathbf{u} \cdot d\mathbf{r} = \int_{-\frac{c}{2}}^{\frac{c}{2}} \frac{\partial \Phi}{\partial y} dx \quad (9)$$

where C is the contour of the airfoil, c is the chord of the airfoil, $\mathbf{u}$ is the velocity field, and Φ is the velocity potential.

The lift coefficient, which is a dimensionless parameter used to quantify the lift on the airfoil, is given by

$$C_L = \frac{\Gamma \rho}{\frac{1}{2} U_\infty^2 c} \quad (10)$$

where ρ is the fluid density and U_∞ is the fluid velocity far from the airfoil.

The drag coefficient, which is a dimensionless parameter used to quantify the drag on the airfoil, is given by

$$C_D = \frac{C_L}{\pi AR} \quad (11)$$

where AR is the aspect ratio of the airfoil, which is defined as the ratio of the square of the chord to the wing area.

In this part we were confronted with the FreeFem simulation and the "thin airfoil theory" computed in a python code. In this code Airfoil geometric proprieties of the airfoil are define by the sum of linearized problem (Camber , Tchikness). We tried to quantify when this approximation is no longer valid.

5.3 Tandem aifoils theory

Knowing the coordinates (x_j^P, z_j^P) of P_j and (x_j^Q, z_j^Q) of Q_j . The coordinates of the points (x_j^V, z_j^V) of V_j , and (x_j^C, z_j^C) of C_j are :

$$\begin{aligned} x_j^C &= x_j^P + \frac{3d}{4} \cos(\theta) \\ z_j^C &= z_j^P - \frac{3d}{4} \sin(\theta) \end{aligned}$$

coordinates (x_j^V, z_j^V) of V_j :

$$\begin{aligned} x_j^V &= x_j^P + \frac{d}{4} \cos(\theta) \\ z_j^V &= z_j^P - \frac{d}{4} \sin(\theta) \end{aligned}$$

And the induced velocity:

$$\begin{aligned} \overrightarrow{V_M(x, z)} &= \frac{\Gamma_j}{2\pi} \vec{e}_y \wedge \frac{\vec{r} - \vec{r}^j}{\|\vec{r} - \vec{r}^j\|^2} = \frac{\Gamma_j}{2\pi} \vec{e}_y \wedge \frac{[x - x_j^V] \vec{e}_x + [z - z_j^V] \vec{e}_z}{(x - x_j^V)^2 + (z - z_j^V)^2} = \\ \frac{\Gamma_j}{2\pi} \left[-\frac{(x - x_j^V) \vec{e}_z}{(x - x_j^V)^2 + (z - z_j^V)^2} + \frac{(z - z_j^V) \vec{e}_x}{(x - x_j^V)^2 + (z - z_j^V)^2} \right] &= u_j \vec{e}_x + w_j \vec{e}_z \end{aligned}$$

The velocity induced by the vortex V_j at point C_i :

$$\begin{aligned} \overrightarrow{V_{Cij}(x, z)} &= \frac{\Gamma_j}{2\pi} \vec{e}_y \wedge \frac{[(x_i^C - x_j^V) \vec{e}_x + (z_i^C - z_j^V) \vec{e}_z]}{R_{ij}^2} = \frac{\Gamma_j}{2\pi} \left[\frac{-(x_i^C - x_j^V) \vec{e}_z}{R_{ij}^2} + \frac{(z_i^C - z_j^V) \vec{e}_x}{R_{ij}^2} \right] \\ &\begin{cases} u_{ij} = \frac{\Gamma_j}{2\pi} \frac{z_i^C - z_j^V}{R_{ij}^2} \\ w_{ij} = \frac{\Gamma_j}{2\pi} \frac{x_i^C - x_j^V}{R_{ij}^2} \end{cases} \end{aligned}$$

The components (U_i, W_i) of the total speed at point C_i :

$$\Rightarrow \begin{cases} U_i = V_\infty \cos(\alpha) + \sum_{j=1}^N \frac{\Gamma_j}{2\pi} \frac{z_i^C - z_j^V}{R_{ij}^2} \\ W_i = V_\infty \sin(\alpha) - \sum_{j=1}^N \frac{\Gamma_j}{2\pi} \frac{x_i^C - x_j^V}{R_{ij}^2} \end{cases}$$

The component of the speed in C_i normal to the segment P_iQ_i :

$$\overrightarrow{V_{Ci}(x, z)} \cdot \vec{n}_i = (U_i \ W_i) \begin{pmatrix} \frac{-\delta z_i}{\sqrt{\delta_x^2 i + \delta_z^2 i}} \\ \frac{\delta x_i}{\sqrt{\delta_x^2 i + \delta_z^2 i}} \end{pmatrix} = 0$$

The circulation can be define by:

$$\begin{aligned} \frac{-U_i \delta z_i}{\sqrt{\delta_x^2 i + \delta_z^2 i}} + \frac{W_i \delta x_i}{\sqrt{\delta_x^2 i + \delta_z^2 i}} &= 0 \\ \Leftrightarrow \\ \frac{-\delta z_i V_\infty \cos(\alpha)}{\sqrt{\delta_x^2 i + \delta_z^2 i}} - \frac{\delta z_i}{\sqrt{\delta_x^2 i + \delta_z^2 i}} \sum_{j=1}^N \frac{\Gamma_j}{2\pi} \frac{z_i^C - z_j^V}{R_{ij}^2} + \frac{V_\infty \sin(\alpha) \delta x_i}{\sqrt{\delta_x^2 i + \delta_z^2 i}} - \frac{\delta x_i}{\sqrt{\delta_x^2 i + \delta_z^2 i}} \sum_{j=1}^N \frac{\Gamma_j}{2\pi} \frac{x_i^C - x_j^V}{R_{ij}^2} &= 0 \\ \sum_{j=1}^N \Gamma_j \frac{(x_i^C - x_j^V) \delta x_i + (z_i^C - z_j^V) \delta z_i}{R_{ij}^2} &= 2\pi(-\delta z_i V_\infty \cos(\alpha) + V_\infty \sin(\alpha) \delta x_i) \\ \Leftrightarrow \\ \sum_{j=1}^N \Gamma_j a_{ij} &= b_i \Leftrightarrow A\Gamma = b \end{aligned} \tag{12}$$

with

$$a_{ij} = \frac{(x_i^C - x_j^V) \delta x_i + (z_i^C - z_j^V) \delta z_i}{R_{ij}^2} \text{ and } b_i = 2\pi V_\infty [-\delta z_i \cos(\alpha) + \delta x_i \sin(\alpha)]$$

And more generally, in the cases of N-plans:

$$\Gamma_{tot} = \sum_{i \geq 0}^N \Gamma_i \frac{1}{1 + 4(\delta i)^2} \tag{13}$$

with $\delta = \frac{d}{c}$, d the distance between the plates, The python code solves this equation 12.